

Bachelor in Computer Applications

Course Title: **Network and Data Communications**

Code No: Semester: **4**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Differentiate between noise and attenuation. A pure ALOHA network transmits 200 bit frames using shared channel with a 50 kbps bandwidth. What is the requirement to make this frame collision free?
2. What is Congestion Control? How can it be handled? Explain acknowledgement policy and discarding policy.
3. What are the practical implications of OSI layer? Define each layer focusing on its functionality and hardware used in each layer.
4. Explain leaky-bucket algorithm with an example.
5. Represent bit sequence 10011001 by the following wave forma. NRZ-Lb. NRZ-I
6. Explain link state routing protocol with an example.
7. Explain recursive resolution. What are its advantages?
8. Calculate the transmitted encoded frame if the message sequence is 1001001 and generator polynomial is $G(X) = x^2 + x + 1$
9. Explain the concept of TDMA with a neat diagram.
10. What is circuit switching? What are its advantages and disadvantages?
11. What is the difference between DNS and DHCP? Explain with example.
12. Write short notesa. Reliable Protocolb. Satellite Network